

MINIMUM WAGE POLICY AND YOUTH EMPLOYMENT: ANALYSING THE 16-19 AGE GROUP

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RESEARCH QUESTION

How does minimum wage policy affect youth employment in the US?

SUB-QUESTIONS

The magnitude of the effect: does a higher minimum wage reduce overall teenage employment, or does it only affect certain subgroups, such as school-going vs. out-of-school individuals?

Are there differential effects by gender, ethnicity, or urban and rural context? Test for heterogeneity.

Do observable impacts occur immediately in the year of the wage hike, or do they manifest over subsequent years?

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OVERVIEW OF MINIMUM WAGE POLICY

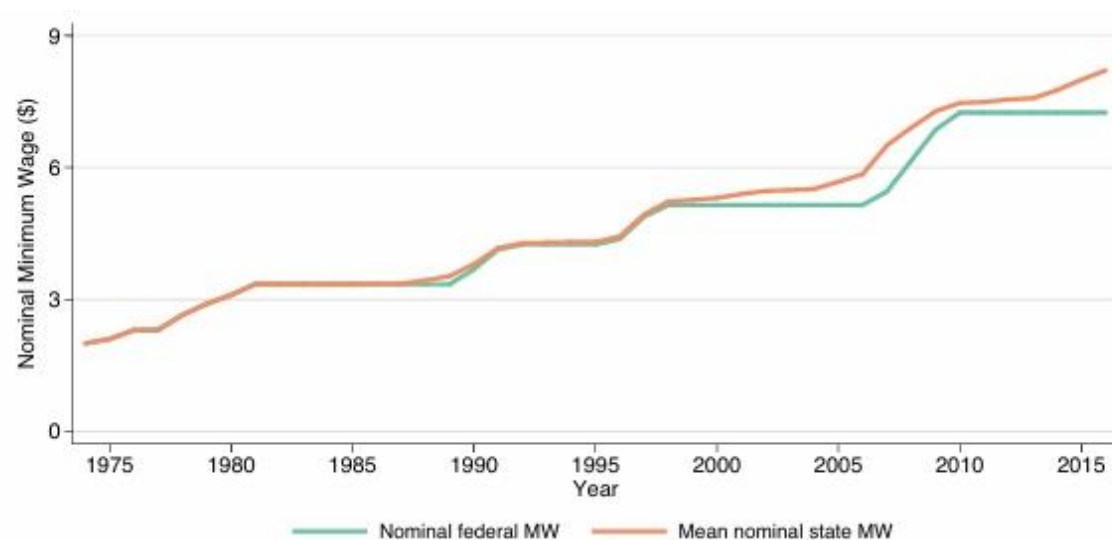
- As of January 1, 2025, the federal minimum wage in the United States remains at \$7.25 per hour, a rate unchanged since July 24, 2009.
- Some states have adopted automatic indexing, tying annual minimum wage increases to inflation.
- Certain categories of workers e.g., tipped workers, student workers and disabled workers in some circumstances can be paid below the standard minimum wage under specific conditions.
- High-cost-of-living regions e.g., Seattle, New York City have pushed for more aggressive minimum wage hikes up to \$15.

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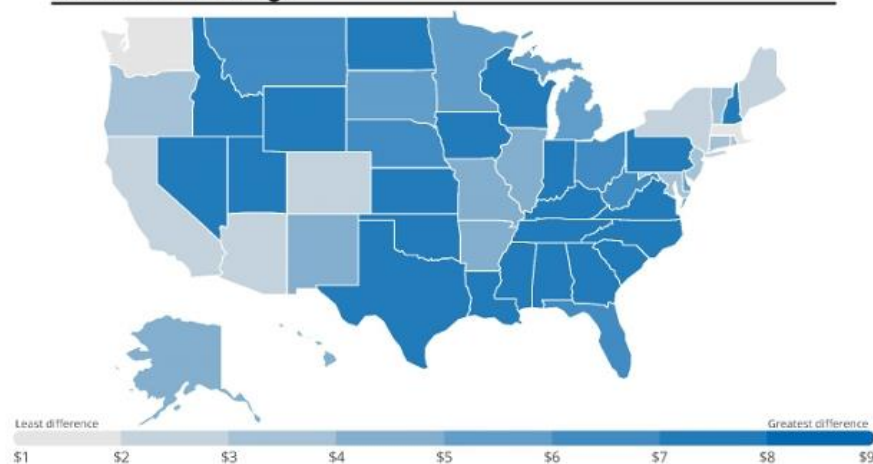
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LITERATURE REVIEW (POLICY)



Difference between Proposed \$15/hr Minimum Wage and Current State Minimum Wage



[†] States that have implemented a phased approach to reach \$15 during the next 3-5 years
Source: [National Conference of State Legislatures](#)

LITERATURE REVIEW

Negative impact on employment rates	No/negligible impact on employment rates	Positive impact on employment rates
Gregory, T. and Zierahn, U., 2022. When the minimum wage really bites hard: The negative spillover effect on high-skilled workers. <i>Journal of Public Economics</i>, 206, p.104582. Wilson, M., 2012. The negative effects of minimum wage laws. <i>Cato Institute, Policy Analysis No. 701</i>. Zavodny, M., 2000. The effect of the minimum wage on employment and hours. <i>Labour Economics</i>, 7(6), pp.729-750. Paun, C.V., Nechita, R., Patruti, A. and Topan, M.V., 2021. The impact of the minimum wage on employment: An EU panel data analysis. <i>Sustainability</i>, 13(16), p.9359.	Schmitt, J., 2015. Explaining the small employment effects of the minimum wage in the United States. <i>Industrial Relations</i>, 54(4), pp.547-581. Gramlich, E.M., Flanagan, R.J. and Wachter, M.L., 1976. Impact of minimum wages on other wages, employment, and family incomes. <i>Brookings Papers on Economic Activity</i>, 1976(2), pp.409-461. Manning, A., 2021. The elusive employment effect of the minimum wage. <i>Journal of Economic Perspectives</i>, 35(1), pp.3-26. Wang, W., Phillips, P.C.B. and Su, L., 2019. The heterogeneous effects of the minimum wage on employment across states. <i>Economics Letters</i>, 174, pp.179-185.	Azar, J., Huet-Vaughn, E., Marinescu, I., Taska, B. and von Wachter, T., 2024. Minimum wage employment effects and labour market concentration. <i>Review of Economic Studies</i>, 91(4), pp.1843-1883. Zipperer, B., 2022. Turnover, prices, and reallocation: Why minimum wages raise the incomes of low-wage workers. <i>Journal of Law and Political Economy</i>, 3(1), pp.160-185. Rao, N. and Risch, M., 2024. Who's afraid of the minimum wage? Measuring the impacts on independent businesses. <i>Working Paper</i>.

LITERATURE REVIEW

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Song, H.J., Yeon, J., Lee, S. and Li, Z., 2022. The effect of an increase in federal minimum wage on the U.S. hotel industry: A difference-in-differences approach. Current Issues in Tourism, 25(6), pp.887–900. Available at: <https://doi.org/10.1080/13683500.2021.1903402> [Accessed 12 March 2025].

Biewen et al., 2022	Cengiz et al., 2019	Song et al., 2022
- First study to use large-scale administrative wage data for a distributional analysis of the German minimum wage. - Application of Distribution Regression Difference-in-Differences (DR-DID) model. - Finds that the minimum wage reduced inequality significantly, but long-term inequality trends were already stabilizing before 2015. - Demonstrates that the German minimum wage had no adverse effects on employment or hours worked. $P(Y_{it} \leq z Bite_g, D_r, D_t, X_{it}) = F(z Bite_g, D_r, D_t, X_{it}) = \alpha_z + D_g \lambda_z + D_t \gamma_z + \beta_z (Bite_g \times D_t) + X_{igt} \delta_z + (X_{igt} \times D_t) \eta_z$	- Examines the impact of minimum wage increases on low-wage employment in the United States using 138 state-level minimum wage changes (1979–2016). - Difference-in-Differences (DID) approach to estimate causal effects. Introduces a bunching estimator to assess how jobs below the new minimum wage shift upwards rather than disappear. - No significant job losses five years after a minimum wage increase. Minimum wage increases redistribute jobs rather than destroy them. $E_{sjt} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} I_{\tau k}^{sjt} + \mu_{sj} + \rho_{jt} + \Omega_{sjt} + u_{sjt}$	- Examines the impact of the 2009 federal minimum wage increase on hotel businesses in the U.S. - Difference-in-Differences (DID) approach comparing NYC (treatment) vs. Washington, D.C. (control) - Negatively affected profitability and total revenue. - Results suggest that hotels adjusted employment structures and benefits to offset wage increases. - Luxury hotels were less affected, while midscale and budget hotels experienced revenue losses. $HotelPerformance_{it} = \beta_0 + \beta_1 City_i + \beta_2 Time_t + \beta_3 (City_i \times Time_t) + \xi_{xit} + \varepsilon_{it}$

METHODOLOGY

(DATA)

The study utilizes data from the Current Population Survey (CPS), covering 1990–2013.

Employment Variables:

- Employed – Fraction of teens 16–19 currently employed.
- Part-Time Work – Fraction employed part time (<30 hours/week).
- Full-Time Work – Fraction employed full time (≥ 30 hours/week).

Industry-Specific Employment:

- Retail Sector Employment – Total, full time, and part time employment in retail.
- Restaurant Sector Employment – Total, full time, and part time employment in restaurants.

Education Status:

- Fraction of teens 16–19 currently enrolled in school.

Demographics:

- Gender
- Age Groups: 16–17, 18–19
- Ethnicity: White non-Hispanics, Minorities

Policy Variables:

- The nominal minimum wage in effect for each state and year.
- CPI – To adjust minimum wages for inflation, ensuring real wage comparisons.

METHODOLOGY

(DID)

REFERENCES

Allegretto, S.A., Dube, A. and Reich, M., 2011. Do minimum wages really reduce teen employment? Accounting for heterogeneity and selectivity in state panel data. *Industrial Relations*, 50(2), pp.205-240. Available at: <https://doi.org/10.1111/j.1468-232X.2011.00634.x> [Accessed 12 March 2025].

Descriptive Statistics

- Summary Tables
- Cross Group Averages
- Time Series Plot

Parallel Trends

- Event-Study Graph

Additional Checks

- Heterogeneous Responses by Industry

Difference-in-Differences (DID) Model

- $EmpOutcome_{s,t} = \alpha + \beta_1 \ln(RealMinWage_{s,t}) + \gamma_s + \lambda_t + \delta_s \cdot t + \varepsilon_{s,t}$
- $EmpOutcome_{s,t}$ is the dependent variable, capturing an employment metric for teenagers 16–19 in state s and year t .
- $\ln(RealMinWage_{s,t})$ is the natural log of the real (inflation-adjusted) minimum wage in state s , year t .
- γ_s is the state fixed effect, controls for all time invariant differences across states.
- λ_t is the year fixed effect, an indicator for each year t .
- $\delta_s \cdot t$ is the State-specific linear time trend



QUESTIONS